

Design Calculation Sheet VBF-T10A-CALC-01

Input

Parameter	Value	Unit	Source
Positive electrode thickness	75.6	um	parameter set OKane2022
Negative electrode thickness	85.2	um	parameter set
Separator thickness	12	um	parameter set
Positive porosity	0.335	-	parameter set
Negative porosity	0.25	-	parameter set
Positive density	3262	kg/m3	parameter set
Negative density	1657	kg/m3	parameter set
Electrode area (h x w)	0.1027	m2	0.065 x 1.58
Nominal capacity	5	Ah	parameter set
Electrolyte conductivity (override)	3	S/m	design (estimate)
Cation transference number (override)	0.6	-	design (estimate)
Cooling coefficient h (override)	80	W/m2/K	design (liquid cooling)

Capacity-Energy

Item	Value	Unit	Formula / source
1C discharge capacity	5.056773637159709	Ah	cell/T10_1c_dfn.json
Discharge energy	18.55037491199281	Wh	integral(V*I)dt / 3600
Cell mass (electrolyte excl.)	43.4545275216	g	sum(layer thickness*(1-porosity)*density*area)
Gravimetric energy density	426.8916490409877	Wh/kg	energy_wh / mass_kg

Mass-NP

Layer	kg/m2	g/cell	
positive_electrode	0.163993788	16.842	
negative_electrode	0.1058823	10.874	
positive_cc	0.0432	4.437	
negative_cc	0.10752	11.042	
separator	0.00252492	0.259	
Total	0.4231210079999999	43.455	
N/P ratio	0.667	-	(0.75*85.2um*33133)/(0.665*75.6um*63104)

Process

Parameter	Positive	Negative	Formula
Areal density (g/m2)	0.16	0.11	thickness*(1-porosity)*density
Compaction density (g/cm3)	2.169	1.243	density*(1-porosity)/1000
Electrolyte fill (g/cell)	6.441		pore volume * 1.2 g/cm3